

Assignment 3

Last date: October 26, 2025

All graphs are assumed to be simple.

1. (i) (3 marks) Let G be a graph with $n \geq 3$ vertices and $\frac{(n-1)(n-2)}{2} + 2$ edges. Show that G has a Hamiltonian cycle.
(Hint: Use following Theorem 1 done in class

Theorem 1: If G is a graph with $n \geq 3$ vertices, and if for every pair of non-adjacent vertices v and w , the following condition holds:

$$\deg(v) + \deg(w) \geq n$$

then G has a Hamiltonian cycle.)

- (ii) (3 marks) In the above Theorem 1, show that if the condition

$$\deg(v) + \deg(w) \geq n$$

is replaced by

$$\deg(v) + \deg(w) \geq n - 1$$

then theorem will no longer be true.

2. (2 marks) Show that an edge is a bridge if and only if it belongs to no cycle.
3. (2 marks) Show that if there is a walk between vertex u and v then there is also a path between u and v .
4. (2 marks) Show that every closed odd walk contains an odd cycle.
5. (2 marks) Let G be a graph such that degree of every vertex is $\geq (n-1)/2$ then show that G is connected.
6. (4 marks) Show that, if n is odd, it is not possible for a knight to visit all the squares of an $n \times n$ chessboard exactly once by knight's moves and return to its starting point.
(Hint: apply one of mid sem question)

7. Let G be a connected graph. A spanning tree of a graph is a subgraph that is a tree and includes all vertices of G . For example, consider a graph G where $V(G) = \{a, b, c\}$ and $E(G) = \{\{a, b\}, \{b, c\}, \{a, c\}\}$. The subgraph G' with $V(G') = V(G)$ and $E(G') = \{\{a, b\}, \{b, c\}\}$ is a spanning tree of G because it is a subgraph of G , contains all vertices of G and it is also a tree.
- (1 marks) Does every connected graph has a spanning tree? Justify.
 - (1 marks) Show by example that there can be more than one spanning tree of the same graph.
 - (4 marks) Show that an edge of G that appears in every spanning tree must be a bridge. That is, show than a bridge edge of the given graph G must be present in all spanning trees of G and conversely if an edge is present in all spanning trees then that edge must be a bridge edge.
8. (4 marks) Let G be a simple graph with vertices $v_1, \dots, v_n$. Let A^k denote the k th power of the adjacency matrix of G under matrix multiplication. Prove that entry i, j of A^k is the number of v_i, v_j -walks of length k in G .

Halls Theorem:

- (2 marks) A graph is called regular if degree of every vertex is exactly same. Prove that a regular bipartite graph $(X \cup Y, E)$ always has a X -perfect matching.
- (4 marks) A **Latin rectangle** is an $r \times n$ matrix with $r \leq n$. Each entry of the matrix has numbers from an alphabet $\{a_1, a_2, \dots, a_n\}$. Think of these as colors – more vibrant that way! The constraint is that any row and any column has *no repeating entry*. So, if we go up a column or left-to-right a row, no color is repeated.

So, for example, the following are examples of Latin rectangles; one is a 2×5 and the other is a 3×5 .

a_1	a_2	a_3	a_4	a_5
a_2	a_3	a_4	a_5	a_1

a_1	a_2	a_3	a_4	a_5
a_3	a_1	a_4	a_5	a_2
a_2	a_5	a_1	a_3	a_4

An $n \times n$ Latin rectangle is called a **Latin square**. A **completion** of an $r \times n$ Latin rectangle is an $n \times n$ Latin square whose first r rows is the Latin rectangle.

Prove that every Latin rectangle be completed?

Matchings and Turan Theorem

- (1 marks) Show a maximal matching that is not maximum.

2. (2 marks) Show a graph with n vertices in which the maximum matching size is 1.
3. (4 marks) Complete this alternate proof of Hall's theorem via augmenting path: Suppose the maximum matching M in a bipartite graph $(X \cup Y, E)$ is not a X -perfect matching. So there exists a vertex x in X which is not covered by M . Consider the set of all vertices $S \subseteq (X \setminus \{x\}) \cup Y$ that is reachable from x via alternating paths (wrt M). Let $S' = (S \cap X) \cup \{x\}$. Then show that $|N(S')| < |S'|$.
4. (2 marks) Tightness of Turan's Theorem: Let n be even. Show a graph G with n vertices and $n^2/4$ edges such that G does not have a triangle.